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Design and Implementation of an Adaptive Accessible AI Chatbot Embedded in Healthcare Patient Portals

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Master's Degree Capstone Project Report

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This report presents the design and implementation of an adaptive accessible AI chatbot embedded within healthcare patient portals to improve accessibility, usability, and patient interaction. The project was developed by Hiral Choksi as part of the Master's Degree Capstone Project in Computer Science and Engineering at the University of Connecticut under the guidance of Instructor Tingting Yu. The proposed system integrates Agentic AI, Retrieval-Augmented Generation (RAG), FHIR interoperability, and accessibility-focused design to provide personalized, context-aware, and supportive healthcare assistance while reducing cognitive burden during patient interactions.

1. Introduction

Digital healthcare systems have significantly transformed the way patients interact with healthcare providers and access medical information. Patient portals such as Epic MyChart allow users to schedule appointments, access laboratory reports, request prescription refills, communicate with healthcare providers, and manage healthcare records online. These systems improve convenience and increase patient engagement; however, many users continue to struggle with complex interfaces and difficult healthcare workflows.

Patients often interact with healthcare portals during emotionally vulnerable situations such as reviewing abnormal medical results, managing chronic illnesses, or seeking urgent healthcare support. During these situations, cognitive burden, stress, and information overload can negatively impact the user experience. Current healthcare portals generally provide the same workflows and interaction patterns to all users regardless of accessibility needs, emotional state, age, technical familiarity, or healthcare literacy.

This project focuses on designing an adaptive and accessible AI-powered healthcare assistant capable of improving healthcare usability and reducing patient burden. By integrating Agentic AI, Retrieval-Augmented Generation (RAG), FHIR interoperability, and accessibility-focused design principles, the system aims to create a more personalized, supportive, and intelligent healthcare experience.

The proposed chatbot provides users with conversational assistance for healthcare-related tasks such as appointment scheduling, understanding laboratory results, medication management, and provider communication. The system also adapts responses based on user needs by simplifying explanations, guiding users through workflows, and supporting accessibility-focused interactions.

The overall goal of this project is to demonstrate how modern AI technologies can enhance healthcare accessibility, improve patient engagement, and create more human-centered healthcare experiences.

2. Problem Statement

Although healthcare patient portals provide important digital healthcare services, many users continue to face significant usability and accessibility challenges while interacting with these systems. Existing portals often rely on complex navigation structures, medical terminology, and static workflows that can become overwhelming for patients during stressful healthcare situations.

Users frequently access healthcare portals while experiencing anxiety, emotional stress, confusion, or urgency. For example, a patient reviewing abnormal laboratory results, an elderly individual managing chronic conditions, or a parent scheduling care for a sick child may struggle to navigate healthcare systems efficiently. Current patient portals generally provide the same interface and interaction flow to all users regardless of their accessibility requirements, emotional state, technological familiarity, or healthcare literacy level.

Additionally, many healthcare systems lack adaptive support mechanisms capable of simplifying medical explanations, guiding users through healthcare processes, or reducing cognitive burden during complex interactions. Traditional healthcare chatbots are often limited to basic rule-based responses or simple FAQ systems that fail to provide contextual understanding and personalized assistance.

These limitations create barriers for:

- Elderly users
- International patients

- Users with disabilities
- Patients with low health literacy
- Individuals experiencing stress or cognitive overload

As healthcare systems continue to expand digitally, there is a growing need for intelligent, adaptive, and accessibility-focused healthcare assistants capable of supporting diverse patient populations through personalized and context-aware interactions.

3. Project Objectives

The primary objective of this project is to design and implement an adaptive accessible AI chatbot that improves healthcare usability, reduces cognitive burden, and enhances patient interaction within healthcare patient portals.

The specific objectives of this project include:

- Designing an adaptive conversational healthcare assistant
- Reducing cognitive load during healthcare interactions
- Providing simplified and personalized healthcare communication
- Supporting accessibility-focused interaction design
- Integrating Retrieval-Augmented Generation (RAG) for contextual healthcare assistance
- Implementing FHIR interoperability for healthcare data integration
- Developing emotionally supportive conversational workflows
- Improving healthcare navigation and patient engagement
- Creating a scalable and modular AI system architecture

The project also aims to demonstrate how modern AI technologies such as Large Language Models, semantic retrieval systems, and agentic workflows can improve digital healthcare experiences while maintaining accessibility, personalization, and usability.

4. Literature Review

Recent advancements in Artificial Intelligence and conversational systems have significantly transformed the capabilities of intelligent digital assistants. Large Language Models (LLMs) such as GPT and Claude have demonstrated advanced natural language understanding, contextual reasoning, and conversational capabilities across multiple domains, including healthcare applications.

Healthcare organizations increasingly explore AI-powered systems to improve patient engagement, automate administrative workflows, and enhance healthcare communication. However, many existing healthcare chatbots remain limited to simple FAQ systems or predefined rule-based interactions that lack contextual awareness, personalization, and adaptive support.

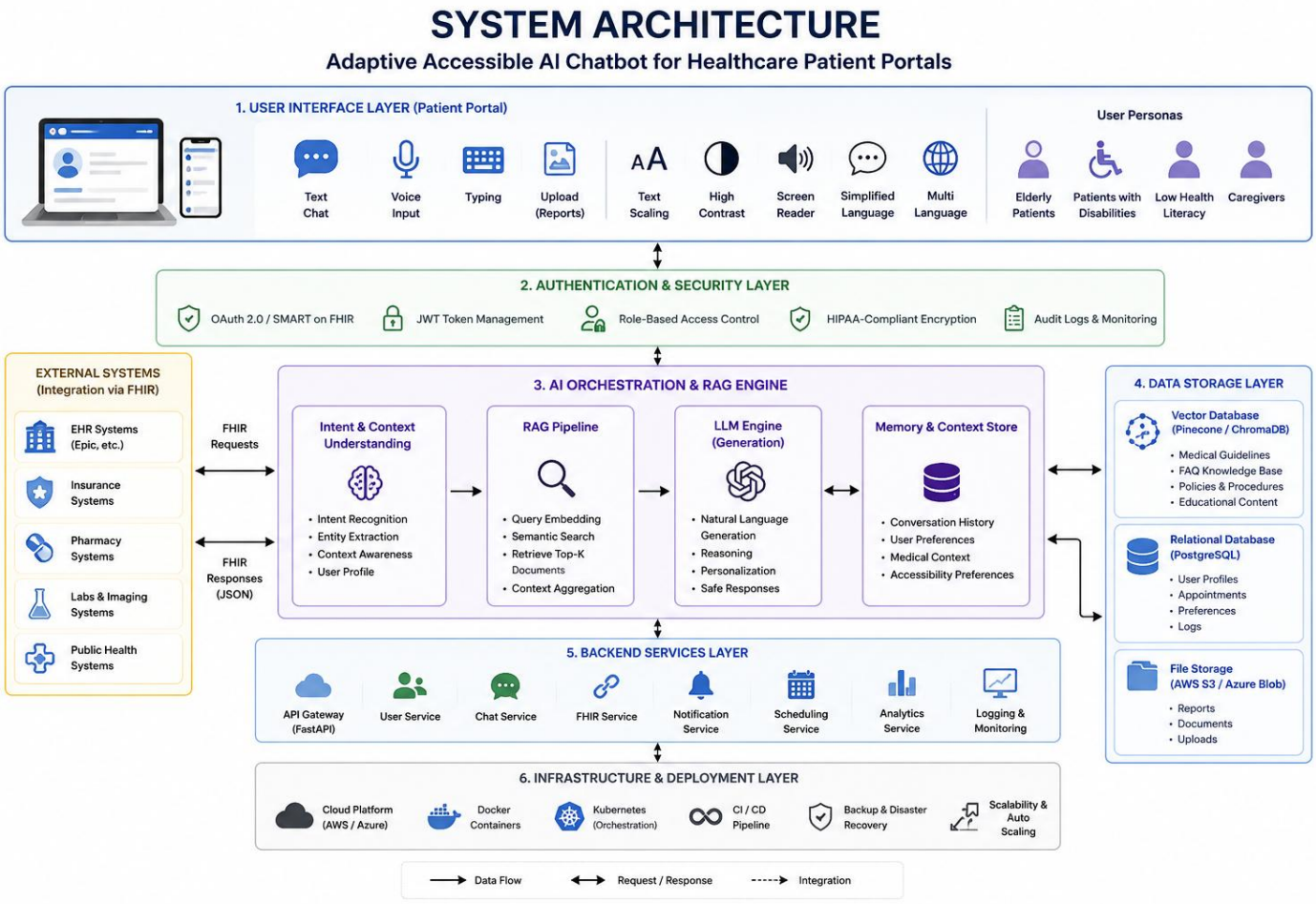
Retrieval-Augmented Generation (RAG) has emerged as an effective methodology for improving the factual accuracy and contextual reliability of AI-generated responses. RAG architectures combine semantic retrieval systems with generative AI models, enabling systems to retrieve relevant contextual information before generating responses. This approach is particularly important in healthcare environments where accurate and trustworthy information is critical.

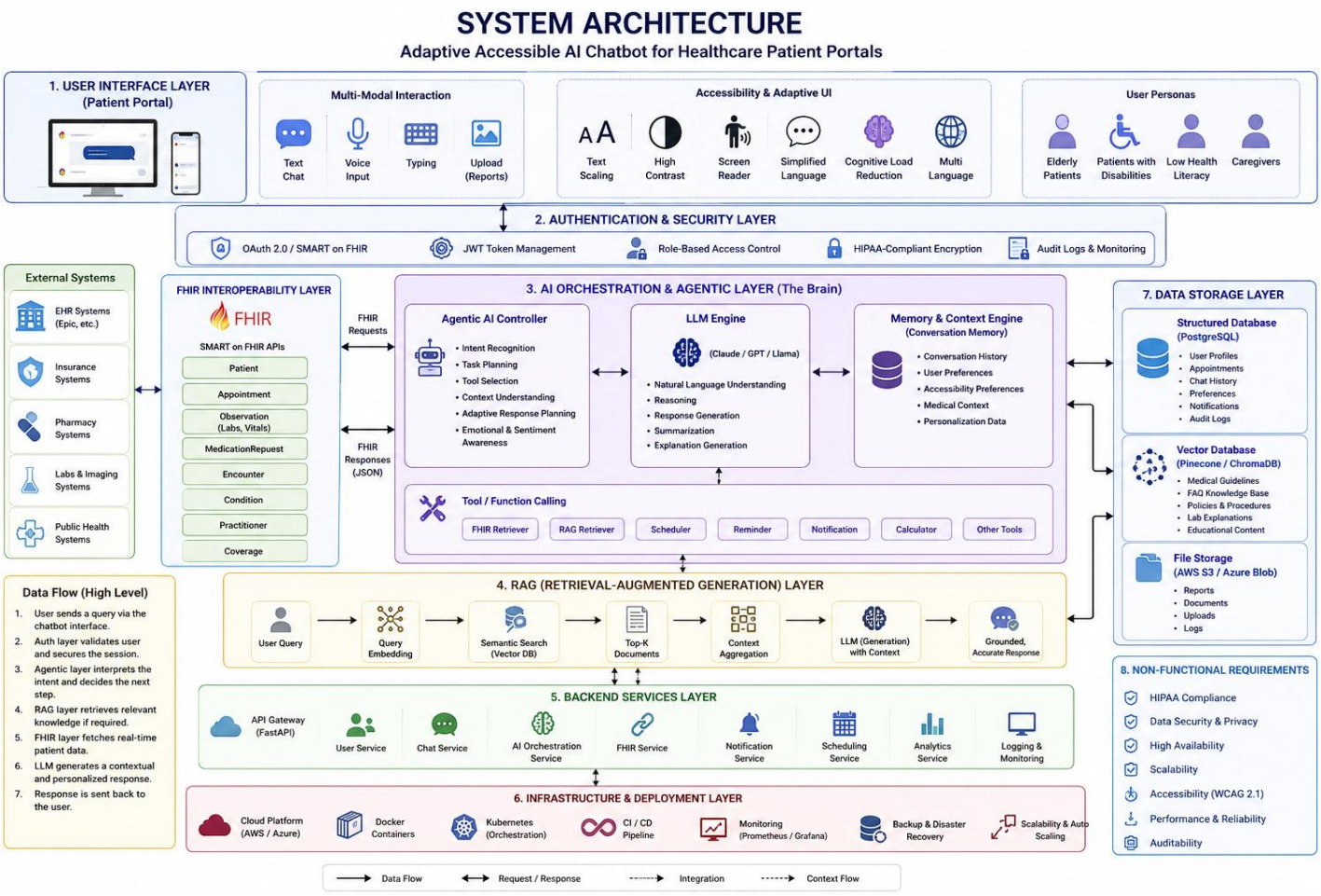
FHIR (Fast Healthcare Interoperability Resources), developed by HL7, is a standardized healthcare interoperability framework that enables secure communication between healthcare systems. FHIR APIs allow healthcare applications to exchange standardized patient information such as appointments, laboratory observations, medications, and healthcare records. Integration of FHIR within AI healthcare assistants enables personalized and context-aware healthcare interactions.

Accessibility and user-centered design also play a critical role in healthcare technology development. Research demonstrates that simplified navigation, plain-language explanations, adaptive workflows, and emotionally supportive communication significantly improve healthcare usability and patient satisfaction. Accessibility-focused healthcare systems can reduce barriers for elderly individuals, users with disabilities, and patients with limited healthcare literacy.

This project combines these research areas by integrating Agentic AI, Retrieval-Augmented Generation, FHIR interoperability, and accessibility-focused design principles into a unified healthcare assistant architecture.

5. System Architecture





5.1 Overview of the Proposed Architecture

The proposed system architecture is designed as a modular, scalable, and accessibility-focused healthcare AI framework capable of supporting adaptive conversational interactions within healthcare patient portals. The architecture integrates Agentic Artificial Intelligence, Retrieval-Augmented Generation (RAG), FHIR interoperability, semantic retrieval systems, contextual memory management, and accessibility-aware interface design to create a personalized healthcare assistance ecosystem.

Unlike traditional healthcare chatbots that rely primarily on static rule-based systems or predefined workflows, the proposed architecture introduces an intelligent orchestration layer capable of dynamically reasoning about user intent, healthcare context, emotional state, and accessibility requirements.

The architecture is organized into seven major layers:

1. Frontend and Accessibility Layer
2. Authentication and Security Layer
3. AI Orchestration Layer
4. Retrieval-Augmented Generation (RAG) Layer
5. FHIR Integration Layer
6. Backend Services Layer

7. Database and Vector Storage Layer

Each layer is designed to function independently while maintaining secure and efficient communication with the overall system.

5.2 High-Level Architectural Workflow

The overall workflow of the proposed healthcare assistant architecture follows the sequence below:

1. The patient securely logs into the healthcare portal.
2. The user submits a healthcare-related query through the conversational interface.
3. The AI orchestration engine analyzes the intent, context, urgency, and accessibility requirements.
4. The system determines whether healthcare retrieval, FHIR access, or contextual memory is required.
5. Relevant healthcare information is retrieved using the RAG pipeline and vector similarity search.
6. FHIR APIs retrieve patient-specific healthcare information such as appointments, medications, or laboratory observations.
7. The Large Language Model generates an adaptive and context-aware response.
8. The frontend interface presents simplified, personalized, and accessibility-focused healthcare guidance.
9. User interactions and preferences are stored securely for future contextual adaptation.

This workflow enables the system to function as an intelligent healthcare assistant rather than a static chatbot.

5.3 Frontend and Accessibility Layer

The frontend layer is responsible for all user-facing interactions and healthcare navigation workflows. The interface is designed using React.js and Tailwind CSS with strong emphasis on accessibility, cognitive burden reduction, and adaptive user experience design.

Traditional healthcare portals often overwhelm users with dense information, difficult navigation structures, and complex medical terminology. Therefore, the frontend architecture prioritizes simplicity, readability, responsiveness, and guided healthcare workflows.

Major Responsibilities of the Frontend Layer

- Conversational healthcare interaction
- Adaptive user interface rendering
- Step-by-step workflow guidance
- Accessibility-focused navigation
- Responsive mobile and desktop layouts
- User preference customization
- Voice and text interaction support

Accessibility Features

The accessibility framework within the frontend layer includes:

- Large-text readability modes
- Simplified medical language settings
- Cognitive load reduction workflows
- Screen-reader compatibility
- Color accessibility compliance
- Reduced interaction complexity
- Emotionally supportive conversational design

The interface dynamically adapts based on user interaction history and accessibility preferences.

5.4 Authentication and Security Layer

Healthcare systems require strict privacy protection, secure communication, and regulatory compliance. The authentication and security layer ensures that all healthcare data interactions remain secure and HIPAA-compliant.

The architecture incorporates OAuth 2.0 authentication and SMART on FHIR authorization standards to enable secure interoperability with healthcare systems.

Security Components

- OAuth2 Authentication
- SMART on FHIR Authorization
- JWT Session Tokens
- Encrypted API Communication
- HIPAA-Compliant Data Handling
- Secure Access Control Policies

This layer prevents unauthorized access to healthcare information while maintaining secure communication between the chatbot and healthcare APIs.

5.5 AI Orchestration Layer

The AI orchestration layer functions as the primary intelligence engine of the system. This layer is responsible for managing conversational reasoning, contextual memory, adaptive response generation, healthcare workflow execution, and decision-making.

Unlike conventional chatbot architecture, the proposed system utilizes Agentic AI principles. The orchestration engine dynamically decides how the system should respond based on healthcare context, emotional indicators, user accessibility needs, and retrieval requirements.

Core Responsibilities of the AI Orchestration Layer

- Intent recognition
- Conversational reasoning
- Accessibility adaptation
- Emotional-context awareness
- Workflow decision management
- Context tracking
- Tool selection and orchestration
- Personalized healthcare assistance

Agentic AI Behavior

The orchestration engine determines:

- Whether the user requires healthcare information retrieval
- Whether FHIR APIs should be queried
- Whether explanations should be simplified
- Whether step-by-step guidance is required
- Whether the user appears stressed or confused
- Which healthcare workflow should be activated

This creates a significantly more adaptive healthcare interaction experience compared to traditional conversational systems.

5.6 Retrieval-Augmented Generation (RAG) Layer

The Retrieval-Augmented Generation layer improves factual reliability and contextual grounding of AI-generated healthcare responses.

Healthcare environments require highly accurate and contextually reliable information. Purely generative AI systems may hallucinate or generate medically inaccurate responses. To reduce this risk, the architecture integrates semantic retrieval mechanisms prior to response generation.

RAG Workflow

1. User query is converted into vector embeddings.
2. Semantic similarity search is performed within the vector database.
3. Relevant healthcare documents and contextual information are retrieved.
4. Retrieved context is appended to the AI prompt.

5. The Large Language Model generates a grounded response.

Healthcare Retrieval Sources

The retrieval layer may access:

- Medical FAQs
- Appointment instructions
- Healthcare policies
- Medication guidance
- Laboratory explanation datasets
- Patient education materials
- Portal navigation workflows

Technologies Used

- LangChain
- Pinecone
- ChromaDB
- Sentence Transformers
- Embedding Models

The RAG architecture significantly improves trustworthiness and contextual reliability within healthcare interactions.

5.7 FHIR Integration Layer

FHIR (Fast Healthcare Interoperability Resources) serves as the interoperability framework connecting the chatbot with healthcare systems.

The FHIR integration layer allows the AI assistant to retrieve and interact with standardized healthcare resources while maintaining secure and compliant communication.

Supported FHIR Resources

- Patient
- Observation
- Appointment
- MedicationRequest
- Encounter
- Practitioner

- DiagnosticReport

Example Workflow

If a user asks:

"Can you explain my recent blood test results?"

The system:

1. Authenticates the user through SMART on FHIR.
2. Retrieves Observation resources through FHIR APIs.
3. Sends relevant healthcare data to the RAG pipeline.
4. Generates simplified explanations through the AI orchestration engine.
5. Presents adaptive healthcare guidance to the patient.

FHIR integration transforms the chatbot from a generalized assistant into a personalized healthcare support system.

5.8 Backend Services Layer

The backend services layer coordinates communication between frontend systems, AI orchestration modules, retrieval pipelines, healthcare APIs, and databases.

The backend architecture is designed using FastAPI-based microservices to support scalability, modularity, and asynchronous processing.

Responsibilities of the Backend Layer

- API routing
- Session management
- AI request orchestration
- FHIR communication handling
- Database communication
- User preference management
- Healthcare workflow execution
- Logging and monitoring

The modular backend structure enables independent scaling of healthcare retrieval systems, conversational services, and API communication pipelines.

5.9 Database and Vector Storage Layer

The architecture incorporates both structured databases and vector storage systems.

Structured Database

PostgreSQL is used for storing:

- User profiles
- Accessibility preferences
- Conversation history
- Healthcare interaction metadata
- Session information
- Portal activity logs

Vector Database

Vector databases such as Pinecone or ChromaDB store semantic embeddings used by the RAG retrieval system.

Stored embeddings include:

- Medical documents
- FAQ datasets
- Healthcare educational materials
- Portal workflow instructions
- Laboratory explanation datasets

The vector retrieval architecture enables semantic similarity search and contextual healthcare retrieval.

5.10 Accessibility-Centered System Design

Accessibility serves as a foundational design principle throughout the entire architecture.

Unlike traditional healthcare systems that treat accessibility as a secondary feature, the proposed architecture integrates accessibility directly into conversational workflows, AI reasoning, interface rendering, and healthcare guidance.

Accessibility-Oriented Design Goals

- Reduce cognitive burden
- Improve healthcare understanding
- Support emotionally vulnerable users
- Simplify healthcare workflows
- Increase healthcare accessibility
- Support diverse patient populations

The system is specifically designed to support:

- Elderly individuals
- International users
- Patients with low healthcare literacy
- Users with disabilities
- Individuals experiencing stress or anxiety

5.11 Scalability and Future Expansion

The proposed architecture is designed to support future scalability and modular expansion.

Potential future enhancements include:

- Emotion-aware conversational AI
- Voice-enabled healthcare interactions
- Wearable healthcare device integration
- Predictive healthcare recommendations
- Real-time multilingual healthcare support
- Integration with real Epic MyChart environments
- Healthcare analytics dashboards
- Multi-agent healthcare reasoning systems

The modular architecture enables independent scaling and future integration of advanced healthcare AI capabilities.

5.12 Architectural Significance

The proposed architecture demonstrates how modern AI systems can move beyond static healthcare chatbots toward adaptive, intelligent, and accessibility-centered healthcare assistants.

By integrating Agentic AI, Retrieval-Augmented Generation, FHIR interoperability, contextual memory systems, and accessibility-focused interaction design, the proposed system addresses several limitations of current healthcare patient portals.

The architecture contributes to emerging research in:

- Human-centered AI
- Accessible healthcare technology
- AI-assisted healthcare workflows
- Adaptive conversational systems

- Healthcare interoperability
- Cognitive burden reduction in digital healthcare environments

The proposed system provides a scalable foundation for future intelligent healthcare ecosystems capable of supporting more inclusive and personalized patient experiences.

3. Methodology

The development of the proposed healthcare chatbot followed a user-centered and accessibility-focused system design methodology. The primary goal of the methodology was to create a secure and adaptive conversational assistant capable of improving healthcare usability while maintaining interoperability, accessibility, and patient privacy.

The development process was divided into multiple phases including system planning, architecture design, frontend implementation, backend orchestration, FHIR integration, accessibility adaptation, and evaluation.

6.1 Requirement Analysis

The first phase involved identifying major usability and accessibility challenges within existing healthcare patient portals. Research was conducted on patient portal workflows, healthcare chatbot systems, accessibility standards, and healthcare interoperability technologies.

Several common issues were identified during the analysis phase:

- Complex healthcare navigation workflows
- High cognitive burden during stressful situations
- Limited accessibility support
- Medical terminology complexity
- Lack of adaptive interaction systems
- Fragmented healthcare information retrieval

Based on these findings, the system requirements were defined to focus on accessibility, adaptive interaction, conversational guidance, and secure healthcare integration.

6.2 System Design Approach

The system was designed using a modular layered architecture to improve scalability, maintainability, and interoperability. Each architectural layer was assigned specific responsibilities to ensure separation of concerns and secure communication between components.

The architecture integrates:

- Frontend conversational interface
- Accessibility adaptation mechanisms

- AI orchestration services
- Retrieval-Augmented Generation (RAG)
- SMART on FHIR authentication
- FHIR-based healthcare interoperability
- Backend workflow management
- Structured and vector databases

This modular approach allows different components of the system to evolve independently while maintaining secure and efficient communication.

6.3 Frontend Development

The frontend interface was designed using React.js and Tailwind CSS with strong emphasis on accessibility and usability. The interface supports adaptive healthcare workflows and conversational healthcare interaction through a clean and simplified user experience.

The chatbot interface includes:

- Conversational messaging interface
- Accessibility preference controls
- Step-by-step interaction guidance
- Confirmation dialogs before actions
- Simplified healthcare explanations
- Responsive mobile and desktop layouts

Accessibility-focused design principles were incorporated throughout the interface to reduce cognitive burden and improve usability for diverse patient populations.

6.4 Backend and AI Orchestration

The backend architecture was implemented using FastAPI-based services to support asynchronous communication, modular API routing, and scalable workflow management.

The AI orchestration layer acts as the intelligence engine of the system and manages:

- User intent recognition
- Conversational context tracking
- Retrieval orchestration
- Workflow execution
- Accessibility adaptation

- Personalized response generation

The orchestration engine dynamically determines whether the system should:

- Retrieve healthcare information
- Access FHIR resources
- Simplify medical explanations
- Activate step-by-step guidance
- Request confirmation before task execution

This enables adaptive conversational behavior instead of static rule-based interactions.

7. Proposed Workflow

The proposed healthcare assistant workflow was designed to support secure, adaptive, and accessibility-focused healthcare interactions within patient portals.

The workflow begins when the user securely logs into the healthcare portal using SMART on FHIR authentication. After authentication, the patient interacts with the chatbot through the conversational interface.

The complete workflow is described below:

1. The patient logs into the healthcare portal securely.
2. The chatbot interface is initialized with the user's accessibility preferences.
3. The user submits a healthcare-related query.
4. The AI orchestration engine analyzes user intent and conversational context.
5. The system determines whether retrieval or FHIR access is required.
6. The Retrieval-Augmented Generation pipeline retrieves contextual healthcare information.
7. FHIR APIs retrieve patient-specific healthcare resources if necessary.
8. The Large Language Model generates a personalized and context-aware response.
9. The frontend interface presents adaptive healthcare guidance.
10. Confirmation prompts are displayed before any sensitive action execution.
11. User interactions are logged securely for future contextual adaptation and auditing.

This workflow enables the chatbot to function as a secure healthcare assistant rather than a static FAQ system.

4.1 Site Identification and Geospatial Data Creation

A verified set of field coordinates was established by manually creating a **custom KML file** using Google Earth. Each of the 16 sites was precisely georeferenced by placing placemarks at their exact latitude and longitude. To

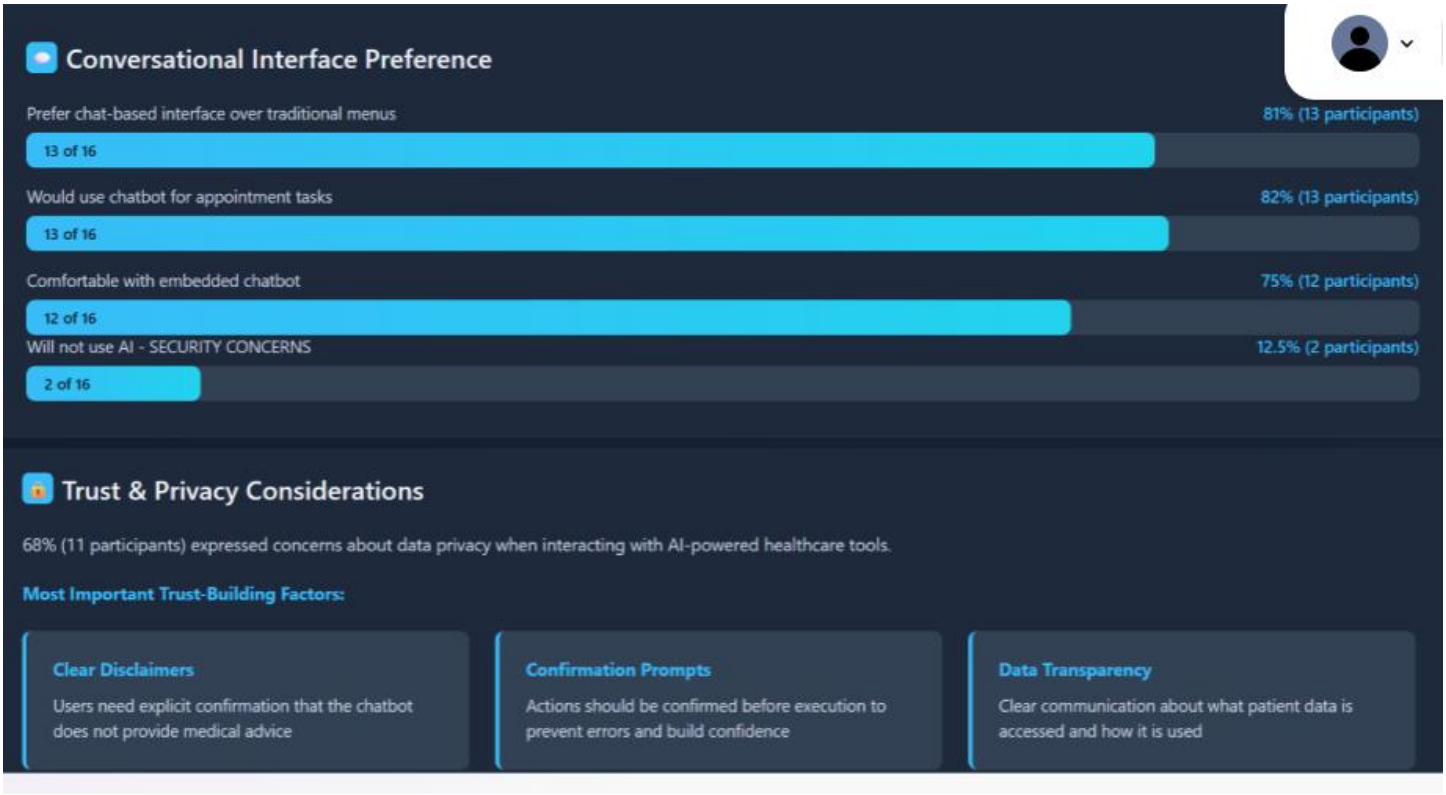
maintain data integrity and organization, sites were grouped into hierarchical folders corresponding to their respective town.

8. User Personas and Use Cases

To evaluate accessibility and usability requirements, multiple healthcare user personas were considered during the system design process



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MAYA PATEL

The Time-Pressed College Student

Routine Checkup Student

- Age: 20
- Occupation: Undergraduate Student
- Life Stage: Full-time student, busy academic schedule

"I just want to book my appointment and move on with my day — not figure out the whole system."

CURRENT BEHAVIOR

- Delays scheduling until urgent
- Clicks through menus quickly
- Screenshots information instead of navigating again
- Sometimes asks friends for help

GOALS

- Schedule routine checkups efficiently
- Quickly access lab results
- Avoid wasting time between lectures
- Feel confident she selected the correct appointment type

EMOTIONAL LAYER

- Mild anxiety when sick
- Frustration when tasks take too long
- Cognitive overload during busy days
- Decision fatigue when choosing appointment types

CHALLENGES

- Too many navigation layers
- Unclear appointment categories
- Medical terminology feels intimidating
- Fear of booking the wrong visit type
- Logs in already stressed or rushed

NEEDS / HOW THE SYSTEM SHOULD HELP

- Guided, step-by-step scheduling
- Plain language explanations
- Confirmation before booking
- Reduced number of clicks
- Clear visual hierarchy

DESCRIPTION

Maya is a sophomore balancing classes, part-time work, and social life. She uses MyChart mainly for routine checkups, vaccination records, and occasional lab results. She logs in between classes and expects tasks to be quick and intuitive.



RACHEL MARTINEZ

Mom Calling for Child's Fever

The Anxious Parent Seeking Urgent Care

- Age: 36
- Occupation: Working Mother
- Life Stage: Managing family + career

"When my child is sick, I don't have mental space to click through five screens."

CURRENT BEHAVIOR

- Double-checks everything
- Considers calling the clinic instead
- Moves quickly but hesitantly
- Refreshes page to re-read instructions

GOALS

- Book the correct pediatric appointment immediately
- Avoid mistakes under pressure
- Understand symptoms and next steps
- Feel reassured she made the right choice

EMOTIONAL LAYER

- High stress
- Protective instinct
- Fear of error
- Emotional urgency

CHALLENGES

- Anxiety affects decision-making
- Multiple appointment types are confusing
- Fear of choosing the wrong provider
- Too many steps during urgent moments
- Worry about missing something important

NEEDS / HOW THE SYSTEM SHOULD HELP

- Direct question-based guidance
- "Is this urgent?" triage direction
- Clear language
- Explicit confirmation before booking
- Immediate reassurance

DESCRIPTION

Rachel uses MyChart to manage her child's healthcare. When her child develops a fever, she logs in urgently, already anxious. She wants clarity, speed, and reassurance.



LINDA THOMPSON

Older Adult Managing Health Issues

The Cautious Older Adult Managing Chronic Conditions

- Age: 67
- Occupation: Retired
- Life Stage: Managing ongoing health concerns

"I don't want to click the wrong thing and create a bigger problem."

CURRENT BEHAVIOR

- Re-reads instructions multiple times
- Writes notes on paper
- Sometimes asks family for assistance
- Moves slowly through tasks

GOALS

- Access health information clearly
- Avoid making mistakes
- Feel confident in decisions
- Maintain independence

EMOTIONAL LAYER

- Cautious
- Slightly intimidated by system
- Concerned about errors
- Mental fatigue when reading long screens

CHALLENGES

- Small text size
- Dense medical terminology
- Fear of clicking wrong button
- Slower navigation pace
- Overwhelmed by too many options

NEEDS / HOW THE SYSTEM SHOULD HELP

- Larger text options
- Simplified language
- Single-question guidance
- Confirmation prompts
- Clear action summaries

DESCRIPTION

Linda uses MyChart to manage prescriptions, review test results, and communicate with providers. She is cautious with technology and wants clarity and control.

11. Challenges and Limitations

- Limited access to real Electronic Health Record (EHR) systems such as Epic MyChart due to healthcare privacy restrictions and institutional access limitations.
- Reliance on sandbox and simulated FHIR environments instead of production healthcare systems.
- Potential risks of inaccurate or misleading AI-generated responses despite the use of Retrieval-Augmented Generation (RAG).
- The chatbot is designed as a supportive healthcare assistant and not as a replacement for professional medical advice or clinical decision-making.
- Healthcare privacy, HIPAA compliance, authentication, and secure communication introduced additional architectural complexity.
- Limited implementation scope due to the capstone timeline, resulting in a Minimum Viable Product (MVP) with a restricted number of workflows.
- Accessibility evaluation and usability testing were conducted on a limited scale and may require broader testing across diverse patient populations.
- Advanced features such as voice interaction, multilingual support, and real-world clinical deployment were not fully implemented within the project scope.
- Ethical concerns related to adaptive AI systems required careful design decisions to avoid hidden profiling or inferred medical assumptions.
- Accessibility adaptation was implemented only through explicit user-selected preferences to maintain transparency and user autonomy.

12. Future Enhancements

The proposed healthcare chatbot architecture provides a scalable foundation for future development and expansion. Although the current system focuses on accessibility-focused conversational assistance and secure healthcare workflows, several advanced capabilities may be incorporated in future iterations.

Potential future enhancements include:

- Voice-enabled healthcare interaction
- Real-time multilingual support
- Emotion-aware conversational assistance
- Wearable healthcare device integration
- Expanded healthcare workflow automation
- Advanced healthcare analytics
- Integration with live healthcare environments such as Epic MyChart
- Personalized healthcare recommendations
- Multi-agent healthcare reasoning systems

Future development may also include larger usability studies, accessibility evaluation across broader patient populations, and real-world healthcare deployment testing.

13. Conclusion

This capstone project presented the design and implementation of an adaptive accessible AI chatbot embedded within healthcare patient portals. The proposed system integrates Agentic Artificial Intelligence, Retrieval-Augmented Generation (RAG), SMART on FHIR interoperability, and accessibility-focused interaction design to improve healthcare usability and patient support.

The project addressed several limitations commonly found in traditional healthcare portals, including complex navigation workflows, cognitive burden, limited accessibility support, and lack of adaptive interaction systems. By incorporating conversational guidance, simplified healthcare communication, step-by-step assistance, and secure healthcare interoperability, the proposed architecture aims to create a more inclusive and user-centered healthcare experience.

The system architecture combines frontend accessibility features, AI orchestration, semantic retrieval systems, FHIR-based healthcare integration, secure backend communication, and modular scalability to support adaptive healthcare workflows. The project also emphasizes responsible AI practices through confirmation-based task execution, preference-based accessibility adaptation, and privacy-aware system design.

Overall, this project demonstrates how modern AI technologies can be integrated with healthcare interoperability standards and accessibility-centered design principles to support more intelligent, secure, and human-centered healthcare interaction experiences.